

Empirical Falsification of Convergence Rate Predictions for the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

The Connes–Consani–Moscovici (CCM) operator construction (arXiv:2511.22755) produces eigenvalues that match imaginary parts of nontrivial Riemann zeta zeros to remarkable precision (55 digits at $(\lambda^2=13, N=120, 200\text{-digit})$; up to 999 digits at $\lambda^2=1000$ in our independent extension). Two papers in the surrounding literature make quantitative predictions about the convergence rate of this construction: CCM Lemma 7.2 conjectures that the smallest Weil-form eigenvector ξ_λ is approximated by a prolate-wave educated guess k_λ at relative rate $O(\lambda^{-2})$, and Śliwiński’s Conjecture 4.1 (arXiv:2601.12133) predicts $\varepsilon(\kappa) \cdot \ln(\kappa)$ approaches a nonzero constant in the regime $\kappa = N = \lambda$. We test both predictions at high precision (primarily 1000 digits via MPFR/GMP). Conjecture 7.2 is empirically falsified: $\text{rel} \times \lambda^2$ grows by approximately 1.116×10^6 across $\lambda^2 \in [13, 1000]$ rather than remaining bounded by a constant. Conjecture 4.1 is empirically unsupported at HP-1000: across $\kappa = N = \lambda \in [50, 500]$, $\varepsilon \cdot \ln \lambda$ ranges over $[7.792, 32.09]$ (full-spectrum, $4.118\times$ spread) and $[1.250, 10.56]$ (trimmed-interior, $8.445\times$ first/last), without approaching a constant, with a sharp upturn at $\kappa = 500$. Theorem 3.1 (Śliwiński’s lower bound) is rigorously satisfied at every configuration with substantial margin ($5.001\times$ – $128.4\times$ ratios). Per-eigenvalue analysis reveals that the construction converges *super-exponentially* on the first eigenvalues — the first Riemann zero is matched to 160.8 digits at $\kappa = 50$ and to 608.4 digits at $\kappa = 200$, with the matching-digit count growing roughly linearly in κ before saturating against the 1000-digit working-precision ceiling at $\kappa = 400$ (accuracy is controlled jointly by λ , N , and working precision; the regime $\kappa = N = \lambda$ scales all three together). The “slow convergence” measured by aggregate error metrics is dominated by high-index eigenvalues that lie at the matrix’s truncation boundary. We further show that the CCM construction is 10^{55} more accurate than naive Mellin truncation (verified at HP-1000), confirming its power is algebraic rather than analytic. The correct convergence question is not “what is the rate?” but “how many eigenvalues converge to full precision as κ grows?”

1 Introduction

The construction of Connes, Consani, and Moscovici [1] proposes a finite-dimensional operator family $D_{\log}^{(\lambda, N)}$ whose spectrum approximates the imaginary parts of the nontrivial Riemann zeros with extraordinary numerical accuracy. The construction has been independently reproduced and extended to 999 matching digits at $(\lambda^2=1000, N=800, \text{HP-1000})$ in our companion paper [3].

A natural question is the rate at which this approximation improves with λ and N . Two recent works in the literature offer specific quantitative predictions:

Conjecture 1 (CCM Lemma 7.2 [1]). *There exists a constant $C > 0$ such that the smallest Weil-form eigenvector ξ_λ is approximated by the prolate-wave educated guess k_λ in relative L^∞ error at rate $\|\xi_\lambda - ck_\lambda\|_\infty / \|\xi_\lambda\|_\infty \leq C \lambda^{-2}$ for some optimal scalar c .*

Conjecture 2 (Śliwiński Conjecture 4.1 [2]). *In the regime $\kappa = N = \lambda$, the limit $\lim_{\kappa \rightarrow \infty} \varepsilon(\kappa) \cdot \ln(\kappa)$ exists and equals a nonzero constant, where $\varepsilon(\lambda, N) = (1/N) \sum_{k=1}^N |\nu_k - \zeta_k|$ is the mean absolute spectral error.*

Both conjectures, if true, would provide a route to proving the Riemann Hypothesis: Conjecture 1 is Step 2 of the proof strategy in [1], and Conjecture 2 would imply $\varepsilon \rightarrow 0$ hence eigenvalue convergence to the Riemann zeros.

In this paper we:

1. Test Conjecture 1 at high precision (primarily 1000 digits) across $\lambda^2 \in \{13, 100, 1000\}$ and find it empirically falsified: the quantity $\text{rel} \times \lambda^2$ grows by a factor of approximately 1.116×10^6 across the tested range;
2. Test Conjecture 2 in its own regime ($\kappa = N = \lambda$) at HP-1000 (3338-bit precision) across $\kappa \in \{50, 100, 150, 200, 300, 400, 500\}$. $\varepsilon(\kappa) \cdot \ln \lambda$ does not approach a constant: full-spectrum values range over $[7.792, 32.09]$ ($4.118\times$ spread), trimmed-interior values over $[1.250, 10.56]$ ($8.445\times$ first/last), with a sharp upturn at $\kappa = 500$. The conjectural limit form is empirically unsupported by this data. Theorem 3.1 (Śliwiński's lower bound) is satisfied at every configuration with substantial margin ($5.001\times$ – $128.4\times$ ratios);
3. Quantify how much of CCM's accuracy is captured by integral transforms of ξ_λ , at HP-1000 (3338-bit) precision: at $\lambda^2 = 13$, naive Mellin truncation gives $O(1)$ error against Riemann zeros while CCM gives 55 matching digits, an improvement factor of 10^{55} ;
4. Argue from this evidence that integral-transform bridges from ξ_λ to the Riemann zeros (Mellin truncation and ξ -weighted variants tested here) fall short of CCM by orders of magnitude, and that the construction's accuracy comes from *algebraic* structure (eigenvalue problem followed by rational-function zeros).

The CCM construction itself is not in question here. Our companion paper [3] verifies its eigenvalues against 2000-digit reference zeros (Step 1 of the CCM proof strategy holds to 999 matching digits at $\lambda^2=1000$, $N=800$, HP-1000). The findings of the present paper concern only the proposed quantitative bridges *from* ξ_λ *to* the Riemann Hypothesis, and the accuracy comparison against alternative integral formulations. All computations are reproducible from the source-available implementation at

<https://github.com/TeamXcelerator/ccm-convergence-rate-falsifications>.

2 Mathematical Setup

We briefly recall the relevant constructions, following the notation of [1] and [2].

2.1 The Weil Quadratic Form and the Eigenvector ξ_λ

Let $\lambda > 1$ and $L = 2 \ln \lambda$. The Weil quadratic form QW_λ is a real symmetric $(2N+1) \times (2N+1)$ matrix with entries

$$\tau_{n,m} = W_{0,2}(V_n, V_m) - W_R(V_n, V_m) - \sum_{\substack{k=p^j \\ k \leq \lambda^2}} \frac{\log p}{\sqrt{k}} q(V_n, V_m)(\log k),$$

where $V_n(u) = (1/\sqrt{L}) e^{2\pi i n \ln(\lambda u)/L}$ forms an orthonormal basis on $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$. The smallest eigenvalue ε_N of τ has eigenvector $\xi = (\xi_{-N}, \dots, \xi_N)$, symmetrized to be even ($\xi_{-j} = \xi_j$) and normalized so $\sum_j \xi_j = \sqrt{L}$. (The forced-even symmetrization is a theoretical guarantee; empirically the natural smallest eigenvector is even without it at every configuration tested above the precision floor — see [3].)

2.2 The Prolate-Wave Educated Guess k_λ

The prolate spheroidal wave operator on $[-\lambda, \lambda]$ is $\text{PW}_\lambda = -\partial_x((\lambda^2 - x^2)\partial_x) + (2\pi\lambda x)^2$. Let $h_{0,\lambda}$ and $h_{4,\lambda}$ be the two even eigenfunctions with 0 and 4 nodes respectively. Define $h_\lambda = h_{4,\lambda} - r h_{0,\lambda}$ with $r = \int h_{4,\lambda} dx / \int h_{0,\lambda} dx$ so that $\int h_\lambda dx = 0$. The educated guess is then

$$k_\lambda(u) := \sqrt{u} \sum_{n=1}^{\lfloor \lambda/u \rfloor} h_\lambda(nu), \quad u \in [\lambda^{-1}, \lambda],$$

i.e. the Eisenstein-like sum map applied to h_λ .

Conjecture 1 predicts that k_λ approximates ξ_λ at relative rate $O(\lambda^{-2})$.

2.3 The Truncated Completed Eta Function $\Lambda_\lambda(s)$

The truncated completed eta function is

$$\Lambda_\lambda(s) := \int_{\lambda^{-1}}^{\lambda} t^{s-1} \omega(t) dt, \quad \omega(t) := \frac{t e^t}{(1 + e^t)^2},$$

and its ξ -weighted variant is

$$G(s) := \int_{\lambda^{-1}}^{\lambda} f_\lambda(u) \omega(u) u^{s-1} du,$$

where $f_\lambda(u) = (1/\sqrt{L}) \sum_n \xi_n e^{2\pi i n \ln(\lambda u)/L}$ is the function reconstructed from ξ_λ . As $\lambda \rightarrow \infty$, $\Lambda_\lambda(s) \rightarrow \Lambda(s) = \Gamma(s+1)\eta(s)$, whose zeros include the Riemann zeros [4].

3 Implementation

Our implementation uses the Xcelerator Toolkit,¹ a publicly available library of high-precision spectral methods for analytic number theory, pinned to a tagged release for reproducibility. The toolkit provides the core numerical primitives (Gauss–Legendre quadrature with disk-cached HP nodes, LU-based inverse iteration, Newton refinement) and the CCM-specific construction, prolate-wave operators, and Mellin transforms. The paper-specific repository contains only the CLI harness and reproduction scripts.

The toolkit is built on Rust (edition 2021), **rug** (v1.30, wrapping MPFR/GMP) for arbitrary-precision arithmetic, **nalgebra** (v0.33) for f64 dense linear algebra, and **rayon** (v1.10) for parallelism. Three subcommands correspond to the three tests:

- **prolate-compare** (Conjecture 1): loads a pre-computed ξ_λ JSON, builds PW_λ via 3-point finite differences on a uniform grid, identifies $h_{0,\lambda}$ and $h_{4,\lambda}$ by parity and node-counting, forms h_λ and applies the $\mathcal{E}$ map, then computes $\|\xi_\lambda - c k_\lambda\|_\infty / \|\xi_\lambda\|_\infty$ on a logarithmically-spaced sample grid in $[\lambda^{-1}, \lambda]$.
- **sliwinski-check** (Conjecture 2): runs the high-precision CCM construction at given (λ, N) configurations, computes the full positive eigenvalue spectrum, and tabulates $\varepsilon(\lambda, N)$ against $1/(4 \ln \lambda)$ and $\varepsilon \cdot \ln \lambda$.
- **mellin-compare** (CCM vs Mellin truncation): scans the critical line $\Re(s) = 1/2$ for zeros of both $\Lambda_\lambda(s)$ and $G(s)$ via Gauss–Legendre quadrature, then bisects in high precision to locate each zero precisely.

The high-precision ξ_λ values are reused from the same runs that produced the eigenvalue verification in our companion paper [3], ensuring consistency between the two papers' empirical foundations.

¹<https://github.com/TeamXcelerator/xcelerator-toolkit>

4 Result 1: CCM Lemma 7.2 is Empirically Falsified

Conjecture 1 asserts that $\text{rel} \times \lambda^2$ should be approximately constant (equal to C) across λ . We measure this quantity at three configurations:

Table 1: Prolate-wave approximation of ξ_λ at HP-1000 working precision (3338-bit, MPFR/GMP). All three ξ_λ values pre-computed at HP-1000 and committed to the data directory; banded tridiagonal LU recovers each prolate eigenvector at HP-1000 in milliseconds.

Configuration	N_{grid}	rel L^∞ error	rel $\times \lambda^2$	λ^{-2} bound
$\lambda^2=13, N=120$	4001	6.887×10^{-5}	8.953×10^{-4}	7.69×10^{-2}
$\lambda^2=100, N=500$	4001	6.697×10^{-4}	6.697×10^{-2}	1.00×10^{-2}
$\lambda^2=100, N=500$	8001	6.470×10^{-4}	6.470×10^{-2}	1.00×10^{-2}
$\lambda^2=1000, N=800$	8001	9.988×10^{-1}	9.988×10^2	1.00×10^{-3}

The growth of $\text{rel} \times \lambda^2$ is dramatic: $8.953 \times 10^{-4} \rightarrow 6.470 \times 10^{-2} \rightarrow 9.988 \times 10^2$ across the three configurations (using the $N_{\text{grid}} = 8001$ row at $\lambda^2=100$ for the apples-to-apples comparison), a factor of approximately 1.116×10^6 . If Conjecture 1 held, this column would be approximately constant.

At $\lambda^2 = 1000$, the residual is 99.9% of ξ_λ itself, with the optimal scalar c at exponential magnitude ($\sim 4.567 \times 10^{501}$). The educated guess has essentially no correlation with the actual eigenvector at large λ .

4.1 Grid Convergence Test

To rule out finite-difference truncation as the cause of the residual, we doubled the grid resolution at $\lambda^2 = 100$:

Table 2: Grid convergence of the prolate-wave residual at $\lambda^2=100$, HP-1000. Halving the FD grid spacing changes the residual by 3.6%, far short of the $4\times$ reduction an $O(h^2)$ method would produce if FD truncation dominated.

	N_{grid}	rel L^∞	change from 4001
4001	6.71253 $\times 10^{-4}$	—	
8001	6.46957 $\times 10^{-4}$	−3.6%	

If the residual were dominated by FD truncation, doubling the grid (halving h) should cut the error by $4\times$ (an $O(h^2)$ method). Instead it changed by 3.6%, ruling out truncation as the driver.

4.2 Interpretation

The prolate operator itself is well-resolved: the smallest prolate eigenvalue at $\lambda^2 = 1000$ is 6282.28, against the asymptotic prediction $2\pi\lambda^2 = 6283.19$ (0.014% relative error). The failure is not in the prolate eigenfunctions, but in the chain $h_{0,\lambda}, h_{4,\lambda} \rightarrow h_\lambda \rightarrow k_\lambda \rightarrow$ comparison with ξ_λ . The educated guess works well at small λ (where PW_λ approaches the harmonic oscillator limit) and breaks down at large λ .

Remark 1. *This finding is consistent with the post-2022 CCM literature trajectory: the authors moved away from the positive prolate spectrum in their 2022 PNAS work (toward the Sonin space), and Śliwiński’s January 2026 paper proves an inverse-logarithmic lower bound via Heisenberg uncertainty that does not rely on prolate waves at all.*

5 Result 2: Śliwiński’s Conjecture 4.1 is Empirically Unsupported at HP-1000

Śliwiński [2] proves Theorem 3.1, the lower bound $\varepsilon(\lambda, N) \geq 1/(4 \ln \lambda)$, via a Heisenberg uncertainty argument: position operator on $L^2([\lambda^{-1}, \lambda], du)$ has support size $2 \ln \lambda$, hence $\sigma_u \leq 2 \ln \lambda$, and $\sigma_u \cdot \sigma_D \geq 1/2$ gives $\sigma_D \geq 1/(4 \ln \lambda)$. Conjecture 2 further claims this bound is asymptotically tight: $\varepsilon(\kappa) \ln(\kappa)$ approaches a nonzero constant.

5.1 Standard CCM Regime ($N \gg \lambda$)

Table 3: Śliwiński’s lower bound applied to our standard CCM configurations, all at HP-1000 (3338-bit) working precision. ε is the mean absolute spectral error against the first N Riemann zeros.

λ	λ^2	N	digits	mean abs ε	$1/(4 \ln \lambda)$	ratio	$\varepsilon \cdot \ln \lambda$
$\sqrt{13}$	13	120	1000	1.063	0.195	$5.454\times$	1.363
10	100	500	1000	1.595	0.109	$14.69\times$	3.673
$\sqrt{1000}$	1000	800	1000	$3.608^\dagger$	0.072	$49.85\times$	12.46

[†]At HP-500 this row previously read $\varepsilon = 5.638$ ($\varepsilon \cdot \ln \lambda = 19.5$). That value was inflated by working-precision saturation of the low-index eigenvalues: at HP-500 the first eigenvalues saturate near 500 matching digits, whereas at HP-1000 they resolve to ~ 999 (Table 6), lowering the mean. The HP-1000 value is the accurate one.

Theorem 3.1 holds with substantial margin ($5\times$ to $50\times$). The conjecture, however, would require $\varepsilon \cdot \ln \lambda$ to be roughly constant; we observe it growing from 1.36 to 12.46 across the three configurations (a factor of 9.14).

5.2 Śliwiński’s Own Regime ($\kappa = N = \lambda$) at HP-1000

This is the regime in which Conjecture 2 was formulated. We test it at 1000-digit working precision (3338 bits, MPFR/GMP) across $\kappa \in \{50, 100, 150, 200, 300, 400, 500\}$. The full-spectrum metric $\bar{\varepsilon} = (1/N) \sum_k |\nu_k - \zeta_k|$ uses all N eigenvalues; the trimmed-interior metric drops the top 20% of edge eigenvalues, which lie at the matrix’s truncation boundary (*cf.* Section 5.4).

Table 4: Śliwiński’s regime $\kappa = N = \lambda$ at HP-1000 (3338-bit precision). Full-spectrum metric uses all N eigenvalues; trim-20% drops the $[0.2N]$ highest-index eigenvalues. “ratio” is the Theorem 3.1 ratio $\varepsilon/[1/(4 \ln \lambda)]$; values > 1 mean the bound is satisfied.

κ	full $\bar{\varepsilon}$	full $\varepsilon \cdot \ln \lambda$	full ratio	trim $\bar{\varepsilon}$	trim $\varepsilon \cdot \ln \lambda$	trim ratio
50	2.938	11.49	$45.97\times$	0.3196	1.250	$5.001\times$
100	1.692	7.792	$31.17\times$	0.9644	4.441	$17.77\times$
150	2.984	14.95	$59.82\times$	0.5834	2.923	$11.69\times$
200	1.579	8.367	$33.47\times$	0.9123	4.834	$19.34\times$
300	2.156	12.30	$49.20\times$	0.8236	4.698	$18.79\times$
400	1.310	7.846	$31.38\times$	0.7553	4.525	$18.10\times$
500	5.163	32.09	$128.4\times$	1.699	10.56	$42.23\times$

Two findings:

1. **Theorem 3.1 holds at every configuration with substantial margin**, in both metrics: ratios from $5.0\times$ ($\kappa=50$ trim) to $128.4\times$ ($\kappa=500$ full). The bound is rigorously satisfied across the entire HP-1000 sweep.

2. **Conjecture 4.1 is empirically unsupported at HP-1000.** The conjecture predicts $\varepsilon(\kappa) \cdot \ln \kappa$ approaches a single nonzero constant. The data does not approach a constant in the tested range. Across $\kappa \in [50, 500]$:

- Full-spectrum: $\varepsilon \cdot \ln \lambda$ ranges over $[7.792, 32.09]$, a $4.118\times$ spread, with first/last ($\kappa=50 \rightarrow 500$) ratio $2.792\times$.
- Trim-20%: ranges over $[1.250, 10.56]$, a $8.445\times$ first/last ratio (both extrema are the endpoints).
- Both metrics show a sharp upturn at $\kappa = 500$: $4.090\times$ jump in the full metric over $\kappa=400$, $2.333\times$ in the trimmed metric.

The $\kappa = 500$ upturn may reflect a structural matrix resonance at this specific (κ, N) pair, or numerical conditioning in the inverse-iteration and Newton refinement stages near the matrix’s truncation boundary; we report both possibilities. Either way, $\varepsilon(\kappa) \cdot \ln \kappa$ does not approach a constant in the tested range, and the conjectural limit form is not supported by this data.

Floor independence. At $\kappa = 400$ and 500 the first eigenvalues saturate at the HP-1000 precision ceiling (~ 999 matching digits; see Table 6), meaning ε_N at these configs is at or below the working-precision floor. This does *not* invalidate the aggregate metric $\varepsilon(\kappa)$: the mean absolute error is dominated by the high-index eigenvalues at the matrix’s truncation boundary, which contribute $O(1)$ error independently of the precision floor. Raising the working precision to HP-2000 would sharpen the first few eigenvalues (from 999 to ~ 2000 digits) but would not materially change $\varepsilon(\kappa)$, because the $O(1)$ truncation-boundary tail still dominates the sum. The upturn at $\kappa = 500$ is therefore structural (more eigenvalues at the boundary contributing larger errors), not a precision-floor artifact.

The companion repository contains the trimmed-metric flag (`--trim-edge-fraction`) used to produce the right-hand columns above, so reviewers can re-run the analysis at any precision and any trim fraction they choose.

5.3 Mean vs. Uniform Error

Śliwiński’s Conjecture 4.1 targets the uniform (pointwise maximum) error $\max_k |\nu_k - \zeta_k|$ rather than the mean $(1/N) \sum_k |\nu_k - \zeta_k|$ used as the primary metric in Section 5; we thank Śliwiński (private communication, 2026) for clarifying this distinction. Our data shows the uniform error also fails to stabilize when multiplied by $\ln^\alpha(\kappa)$ for any α :

Table 5: Uniform error $\times \ln^\alpha(\kappa)$ at HP-1000 (3338-bit precision). If the uniform error decayed as $1/\ln^\alpha$, one column would be approximately constant. None do — the values oscillate over orders of magnitude across κ , with a sharp upturn at $\kappa = 500$.

κ	uniform err	unif $\cdot \ln^1$	unif $\cdot \ln^2$	unif $\cdot \ln^3$
50	66.05	258.4	1011	3955
100	38.87	179.0	824.3	3796
150	154.5	774.3	3880	19440
200	60.59	321.0	1701	9013
300	139.5	795.8	4539	25890
400	52.70	315.7	1892	11340
500	629.1	3910	24300	151000

No column stabilizes. The uniform error is dominated by the *last* eigenvalue ($k = N$), which stays at $\lesssim 3$ matching digits regardless of κ (cf. Table 6, last column).

Śliwiński (private communication) further suggested a softer extended-form conjecture — $1/\ln^\alpha(\kappa)$ for some $\alpha > 1$, rather than the original $\alpha = 1$. Table 5 tests this directly: if the uniform error decayed as $1/\ln^\alpha(\kappa)$ for any fixed $\alpha \in \{1, 2, 3\}$, the corresponding column would be approximately constant across κ . None of the columns $\alpha \in \{1, 2, 3\}$ stabilize at HP-1000 in the tested range $\kappa \in [50, 500]$; the values vary by factors of $\sim 16\times$ ($\alpha = 0$, raw uniform) to $\sim 40\times$ ($\alpha = 3$) within each column, and the sharp upturn at $\kappa = 500$ ($\sim 12\times$ over $\kappa = 400$) is preserved in every column independent of α . The extended-form conjecture is therefore also unsupported at HP-1000 in this regime, with the same caveat as before: the conjecture is asymptotic and our test is finite.

5.4 Per-Eigenvalue Convergence Profile at HP-1000

The HP-1000 spectrum exhibits a sharp split between converged and unconverged eigenvalues:

Table 6: Matching digits per eigenvalue index at HP-1000. Each entry is $-\log_{10} |\nu_k - \zeta_k|/|\zeta_k|$ computed in HP arithmetic. The first eigenvalues converge super-exponentially; the last eigenvalue ($k = N$) remains at $\lesssim 3$ matching digits regardless of κ .

κ	$k=1$	$k=5$	$k=10$	$k=20$	$k=N$
50	160.8	126.5	61.2	15.5	2.2
100	315.1	290.3	254.0	135.7	2.8
150	463.3	441.8	412.8	340.7	2.6
200	608.4	588.9	563.3	504.9	0.8
300	893.0	875.7	853.9	806.9	1.3
400	999.4	1001	1000	988.8	2.4
500	999.4	1001	1000	1000	1.9

The first eigenvalue’s matching-digit count grows roughly linearly with κ through $\kappa = 300$ ($160 \rightarrow 315 \rightarrow 463 \rightarrow 608 \rightarrow 893$, an additive trend of ~ 140 – 150 digits per $\Delta\kappa = 50$), then saturates against the 1000-digit working-precision ceiling at $\kappa = 400$ and beyond ($999 \rightarrow 999$). At $\kappa = 200$ the first 20 eigenvalues all match to > 500 digits; at $\kappa = 300$, > 800 ; at $\kappa = 400$, > 988 ; at $\kappa = 500$, all of the first 20 are at the precision ceiling. The last eigenvalue ($k = N$) remains at $\lesssim 3$ matching digits regardless of κ , confirming that the aggregate metrics in Table 4 are dominated by the unconverged tail at the truncation boundary, not by any intrinsic convergence rate of the construction.

Remark 2. *The $\sqrt{N}$ growth of mean error visible in Table 4 (full-spectrum mean rising from 1.69 at $\kappa=100$ to 5.16 at $\kappa=500$, roughly $\sqrt{5}\times$ for a $5\times$ increase in N) is consistent with a central limit theorem effect: if the first $O(1)$ eigenvalues converge to full precision and the remaining $N - O(1)$ contribute $O(1)$ error each, the mean over N terms grows as $\sqrt{N}$ by the CLT applied to the GUE-distributed zero spacings. This connection was noted by Śliwiński (private communication, 2026).*

6 Result 3: CCM is 10^{55} More Accurate than Naive Mellin Truncation

If a substantial fraction of CCM’s accuracy could be captured by an integral transform of ξ_λ , that would constitute a candidate bridge to RH. We test this directly.

6.1 Naive Mellin Truncation

We computed zeros of $\Lambda_\lambda(s)$ on the critical line $\Re(s) = 1/2$ via Gauss–Legendre quadrature with 500 points at HP-1000 precision (3338 bits), followed by HP bisection.

Table 7: Zeros of the truncated completed eta function $\Lambda_\lambda(s)$ on the critical line, $\lambda = \sqrt{13}$, HP-1000.

k	Λ_λ zero	Riemann zero ζ_k	difference
1	14.797	14.135	0.6622
2	22.112	21.022	1.090
3	24.555	25.011	0.4557
4	29.445	30.425	0.9803
5	31.890	32.935	1.045

Table 8: Same comparison at $\lambda = 10$ ($\lambda^2 = 100$).

k	Λ_λ zero	Riemann zero ζ_k	difference
1	13.719	14.135	0.416
2	20.518	21.022	0.504
3	24.603	25.011	0.408
4	30.053	30.425	0.372
5	32.779	32.935	0.156

In both cases the absolute error is $O(0.1\text{--}1.0)$, i.e. *less than one matching digit*. The CCM construction at the same (λ, N) matches Riemann zeros to 55 digits at $\lambda^2 = 13$, 460 digits at $\lambda^2 = 100$, and 999 digits at $\lambda^2 = 1000$ [3]. The improvement factor is therefore $\sim 10^{55}$, $\sim 10^{460}$, and $\sim 10^{999}$ across the three configurations.

6.2 ξ -Weighted Mellin Transform

If ξ_λ encodes essential information about Riemann zeros, weighting the Mellin integrand by $f_\lambda(u)$ might recover some of the CCM accuracy. We test this with the ξ -weighted variant $G(s)$ at $\lambda^2 = 13$ at HP-1000 (3338-bit) precision over the scan range $\Im(s) \in [5, 55]$:

Table 9: ξ -weighted Mellin $G(s)$ vs naive Λ_λ at $\lambda = \sqrt{13}$, $N = 120$, full HP-1000 arithmetic (3338-bit precision, 500-point Gauss–Legendre quadrature, 5000 scan points). Eleven Riemann zeros lie in the scan range.

k	Λ_λ zero	$G(s)$ zero	Riemann zero	Λ_λ error	G error	improvement
1	14.797	13.930	14.135	0.6622	0.2045	3.238×
2	22.112	20.815	21.022	1.090	0.2067	5.275×
3	24.555	25.152	25.011	0.4557	0.1411	3.230×
4	29.445	30.336	30.425	0.9803	0.08852	11.07×
5	31.890	33.399	32.935	1.045	0.4636	2.253×
6	36.784	37.725	37.586	0.8026	0.1389	5.779×
7	41.678	41.322	40.919	0.7595	0.4034	1.883×
8	44.126	43.829	43.327	0.7989	0.5018	1.592×
9	49.022	48.162	48.005	1.017	0.1570	6.479×
10	49.022	50.585	49.774	0.7519	0.8107	0.9274×
11	53.919	53.480	52.970	0.9482	0.4384	2.163×

Mean G error: 0.3231. Mean Λ_λ error: 0.8464. Mean improvement factor: **2.619** $\times$.

Remark 3 (Zero counts on the critical line). *The Λ_λ scan over $\Im(s) \in [5, 55]$ at $\lambda^2 = 13$ finds 21 zeros, while the ξ -weighted $G(s)$ scan over the same range finds 11 zeros — exactly the count of Riemann zeros in the range. The unweighted truncation introduces approximately ten spurious zeros that are not present in the weighted variant. This is consistent with ξ -weighting suppressing oscillatory artefacts of the bare integrand $\omega(t)$ near the truncation endpoints, while the 11 zeros that survive align (modulo the residual $O(0.1\text{--}0.5)$ errors above) with the actual Riemann zeros.*

The ξ -weighting helps modestly but does not approach the 10^{55} accuracy of the full CCM construction. The HP-1000 measurement reproduces the previously reported 2.619 $\times$ improvement to four significant figures, confirming that the residual $O(0.1\text{--}0.5)$ errors are mathematical in nature, not numerical artefacts: at higher precision the same answer emerges.

6.3 Interpretation

The 10^{55} improvement factor of CCM over Mellin truncation quantifies what the Weil quadratic form’s algebraic structure contributes *beyond* integration:

- The eigenvalue problem $\tau\xi = \varepsilon_N\xi$ extracts the smallest eigenpair with full precision available.
- The rational function $R(z) = \xi_0 + 2z \sum_j \xi_j / (z - j^2)$ extracts zeros via an algebraic mechanism whose accuracy is bounded only by ε_N .
- Neither operation has an integral-transform analogue that captures comparable information from ξ_λ alone.

The implication for the bridge problem is sharp: no integral transform of ξ_λ tested here approaches the accuracy CCM exhibits, and the structural argument above (eigenvalues together with rational-function zeros lose information under integration) suggests this generalizes. The bridge must engage with the algebraic mechanism (eigenvalue problem followed by rational-function zeros).

7 Discussion

7.1 Status of the Two Quantitative Bridges

Conjecture 1 proposed an analytic bridge from ξ_λ to the Riemann zeros via prolate-wave approximation at rate $O(\lambda^{-2})$. Our high-precision data falsifies the predicted rate: $\text{rel} \times \lambda^2$ grows by $\sim 1.116 \times 10^6$ across $\lambda^2 \in [13, 1000]$. At $\lambda^2 = 1000$ the residual is 99.9% of ξ_λ itself, with the optimal scalar c at exponential magnitude ($\sim 4.567 \times 10^{501}$); the educated guess has essentially no correlation with the actual eigenvector at large λ .

Conjecture 2 proposed an asymptotic rate $\varepsilon \sim 1/\ln \lambda$ in the regime $\kappa = N = \lambda$. At HP-1000 (3338-bit) across $\kappa \in [50, 500]$ (Section 5.2), $\varepsilon \cdot \ln \lambda$ ranges over $[7.792, 32.09]$ (full-spectrum, 4.118 $\times$ spread) and $[1.250, 10.56]$ (trimmed-interior, 8.445 $\times$ first/last), with a sharp upturn at $\kappa = 500$ in both metrics. Neither trajectory is consistent with the conjectural limit $\varepsilon(\kappa) \cdot \ln(\kappa) \rightarrow C$. Theorem 3.1 (Śliwiński’s lower bound) holds with substantial margin at all configurations (5.001 $\times$ –128.4 $\times$ ratios). Whether the $\kappa = 500$ upturn reflects a structural matrix resonance or numerical conditioning at the matrix’s truncation boundary is under investigation; either way, the data does not support the predicted asymptotic constancy in the tested range.

7.2 The Algebraic Picture

The per-eigenvalue analysis (Table 6) reveals that the CCM construction does not have a convergence rate problem for the first eigenvalues. In the Śliwiński regime ($\kappa = N = \lambda$), the first eigenvalue matches the first Riemann zero to 160.8 *digits* at $\kappa = 50$ and 608.4 *digits* at $\kappa = 200$, with the matching-digit count growing roughly linearly in κ through $\kappa = 300$ (additive trend $\sim 140\text{--}150$ digits per $\Delta\kappa = 50$) before saturating against the 1000-digit working-precision ceiling at $\kappa = 400$ and beyond. At $\kappa = 200$, the first 20 eigenvalues all match Riemann zeros to >500 digits; at $\kappa = 300$, >800 ; at $\kappa = 400$ and beyond, the first 20 are at the precision ceiling.

The construction’s accuracy on these eigenvalues is limited only by arithmetic precision, not by any intrinsic convergence rate. No rate of the form $1/\ln^\alpha(\kappa)$ describes this behavior — it is super-exponential convergence that immediately saturates at whatever precision is available.

The “slow convergence” measured by the mean and uniform errors comes entirely from the *last* eigenvalue ($k = N$), which remains at $\sim 1\text{--}3$ matching digits regardless of κ even at HP-1000 working precision. This is a structural property of the matrix’s truncation boundary, not of the working precision. The correct question is not “what is the convergence rate?” but rather “how many eigenvalues converge to full precision as κ grows?” The HP-1000 per-eigenvalue table shows this count growing rapidly with κ ; characterizing this growth precisely — and proving it unbounded — remains open.

Combined with the 10^{55} improvement of CCM over naive Mellin truncation, the picture that emerges is:

- The CCM construction’s accuracy is *algebraic*, not analytic. It comes from solving an eigenvalue problem on a specific symmetric matrix, then finding zeros of a rational function determined by the eigenvector.
- Neither step has an integral-transform analogue that preserves the full information. Hence no Mellin-style bridge can recover comparable accuracy.
- The candidates studied in the literature (prolate approximation, inverse-logarithmic rate, naive or ξ -weighted Mellin) all fall short by orders of magnitude when tested at high precision.

The construction works (Step 1 of the CCM proof strategy is verified to 999 digits at $\lambda^2=1000$, $N=800$, HP-1000 in [3]) but the quantitative bridges proposed so far do not. A successful proof of RH via this route requires a new framework that engages with the algebraic structure directly and predicts a rate consistent with high-precision data.

7.3 What Remains

Two directions remain open:

1. A different rate prediction in the $\kappa = N = \lambda$ regime, consistent with our $\varepsilon \sim \sqrt{N}$ observation, would refine Śliwiński’s framework.
2. A new bridge construction that does *not* go through prolate waves or Mellin transforms could complete the CCM proof strategy. Candidate frameworks in the literature (CCM 2022 PNAS Sonin space, Yakaboylu’s Hilbert–Pólya Hamiltonian) share operators with CCM 2025 but do not yet provide explicit bridges from ξ_λ .

8 Conclusion

We have empirically tested two published quantitative predictions about the CCM zeta spectral triple’s convergence rate at high precision. CCM Lemma 7.2 fails by a factor of approximately

1.116×10^6 in $\text{rel} \times \lambda^2$ across $\lambda^2 \in [13, 1000]$. Śliwiński’s Conjecture 4.1 is empirically unsupported at HP-1000: across $\kappa = N = \lambda \in [50, 500]$, $\varepsilon \cdot \ln \lambda$ ranges over $[7.792, 32.09]$ (full-spectrum) and $[1.250, 10.56]$ (trimmed-interior, an $8.445 \times$ first/last change), with a sharp upturn at $\kappa = 500$, neither metric approaching a single nonzero constant. Theorem 3.1 (Śliwiński’s lower bound) holds with substantial margin throughout. The CCM construction is approximately 10^{55} more accurate than naive Mellin truncation, and ξ -weighting the Mellin integrand recovers only a $2.6 \times$ improvement (verified at HP-1000).

The per-eigenvalue analysis at HP-1000 reveals a more nuanced picture than simple falsification. The first eigenvalues converge *super-exponentially* — the first Riemann zero is matched to 160.8 digits at $\kappa = 50$ and to 608.4 digits at $\kappa = 200$, with linear growth in κ through $\kappa = 300$ before saturating against the 1000-digit working-precision ceiling at $\kappa = 400$. The aggregate error metrics (mean, uniform) are dominated by high-index eigenvalues lying at the matrix’s truncation boundary. Even at HP-1000 the last eigenvalue ($k = N$) remains at $\sim 1\text{--}3$ matching digits regardless of κ , indicating a structural limit at the boundary rather than a numerical one. The $\sqrt{N}$ growth of the mean error visible in our data is consistent with a CLT effect on GUE-distributed zero spacings.

The construction itself is not in question — it works extraordinarily well on the first eigenvalues. What is in question is the route from ξ_λ to the Riemann Hypothesis. The correct convergence question is: how does the number of fully-converged eigenvalues grow with κ ? Our HP-1000 data shows the first ~ 20 eigenvalues all matching Riemann zeros to >500 digits at $\kappa = 200$, with the count of fully-converged eigenvalues growing rapidly with κ ; characterizing this growth precisely — and proving it unbounded — remains open.

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Data and Code Availability

The complete source code, reference data, and reproduction scripts are available at:

<https://github.com/TeamXcelerator/ccm-convergence-rate-falsifications>

The shared mathematical library (Xcelerator Toolkit) is at:

<https://github.com/TeamXcelerator/xcelerator-toolkit>

References

- [1] A. Connes, C. Consani, H. Moscovici, *Zeta Spectral Triples*, arXiv:2511.22755 (2025).

- [2] D. Śliwiński, *Spectral Analysis of the $D_{\log}^{(\lambda, N)}$ Operators*, arXiv:2601.12133 (2026).
- [3] R. Andrews, Jr., *Independent Reproduction and Convergence Analysis of the Connes–Consani–Moscovici Zeta Spectral Triple*. GitHub: TeamXcelerator/ccm-reproduction-and-convergence (2026).
- [4] E. Yakaboylu, *Hamiltonian for the Hilbert–Pólya Conjecture*, J. Phys. A: Math. Theor. 57, 235204 (2024). arXiv:2309.00405; see also arXiv:2408.15135.
- [5] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, http://www.dtc.umn.edu/~odlyzko/zeta_tables/.
- [6] The PARI Group, *PARI/GP version 2.15*, Université de Bordeaux, 2023. <http://pari.math.u-bordeaux.fr/>.